

ATVB IN FOCUS: Preeclampsia

Guest Editor: Jessica Faulkner

Endothelial Dysfunction in Preeclampsia: Focus on the Uteroplacental Circulation

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ABSTRACT: Preeclampsia is a pregnancy disorder characterized by de novo maternal hypertension, end-organ damage, and fetal complications. It is also associated with increased long-term cardiovascular risk for both the mother and offspring. A general lack of understanding of this disease prevents effective treatment and prevention, with delivery of the fetoplacental unit being the current standard of care for severe cases. A hallmark characteristic of the maternal syndrome is endothelial dysfunction in both reproductive and nonreproductive vascular beds. As the uterine vascular bed serves as the interface between the maternal and fetal circulation, it is pivotal in the cardiovascular function of both mother and fetus. In this review, we highlight uterine artery endothelial maladaptation as a central feature of preeclampsia and discuss its anatomic, cellular, and circulating correlates.

Key Words: cardiovascular diseases ■ endothelium, vascular ■ metabolic syndrome ■ placenta ■ risk factors

Preeclampsia is a multifactorial hypertensive disorder of pregnancy, impacting 2% to 8% of pregnancies.^{1,2} While multiple mechanisms have been proposed as potential drivers and features of preeclampsia, maternal endothelial cell activation or injury is a major mechanistic feature of the disease.

In preeclampsia, endothelial dysfunction encompasses reduced endothelium-dependent relaxation (NO and PGI₂ [prostacyclin 2]), increased vasoconstriction (thromboxane A₂ and ET-1 [endothelin-1]), oxidative stress, a proadhesive and prothrombotic phenotype, and increased permeability. Because endothelial dysfunction can persist after delivery, preeclampsia confers increased long-term maternal cardiovascular risk (2–4-fold in otherwise healthy women and up to 12-fold with recurrent preeclampsia and metabolic syndrome).³

Although endothelial dysfunction has systemic consequences, the uterine circulation has features that may increase vulnerability in complicated pregnancies, including extensive remodeling, altered shear-sensing, hormone responsiveness, and direct crosstalk with immune and trophoblast cells.^{4–6} These unique characteristics contribute to increases in uterine artery blood flow >10-fold

in healthy human pregnancies, while dysfunction of the endothelium contributes to preeclampsia.^{4,7,8} Preeclampsia is a pregnancy-associated syndrome, yet it often occurs in individuals with preexisting (sometimes subclinical) vascular and cardiometabolic vulnerability⁹ that may limit the capacity for normal uteroplacental vascular adaptation. Dysfunction of the uteroplacental vascular endothelium may initiate a feed-forward axis in which placental stress causes the release of vasoactive, proinflammatory, and antiangiogenic molecules into the maternal circulation, where they act on the maternal vascular endothelium to propagate systemic endothelial dysfunction.¹⁰

In this review, we posit that the maternal vascular syndrome of preeclampsia is endothelial in nature, and the uterine vasculature exhibits distinctive endothelial phenotypic changes that contribute to preeclampsia pathology and perinatal outcomes. We summarize the anatomy and function of the uterine circulation, discuss trophoblast-endothelial-immune interactions in the process of uterine spiral artery remodeling, and review circulating biomarkers and understudied mechanisms relevant to uteroplacental endothelial dysfunction to define priorities for future research.

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Nonstandard Abbreviations and Acronyms

ET-1	endothelin-1
EV	extracellular vesicle
Flt-1	fms-like tyrosine kinase 1
IKCa	intermediate-conductance calcium-activated potassium channel
IL	interleukin
iNOS	inducible nitric oxide synthase
MMP	matrix metalloproteinase
PGI₂	prostacyclin 2
PIGF	placental growth factor
sFlt	soluble fms-like tyrosine kinase
SKCa	small-conductance calcium-activated potassium channel
TGF-β	transforming growth factor-beta
TLR	toll-like receptor
TNF-α	tumor necrosis factor-alpha
VCAM-1	vascular cell adhesion molecule-1
VEGF	vascular endothelial growth factor
VEGFR-1	vascular endothelial growth factor receptor-1

ANATOMY AND PHYSIOLOGY OF THE UTERINE CIRCULATION

Both human and animal models are used to study the uterine circulation. Basic uterine vascular patterns are conserved across mammalian species, along with increases in uterine blood flow with pregnancy. Key differences exist between species, including uterine anatomy (eg, bicornuate uterus in rodents versus a single cavity in humans), gestational length and litter size, and placentation mechanisms, and these factors should be considered when extrapolating mechanistic insights to human pregnancy.¹¹

In humans, the maternal uterine circulation originates with the main uterine artery that runs along either side of a single uterine cavity and branches into arcuate arteries within the myometrium and endometrium.¹² These give rise to spiral arteries that form the maternal-placental interface and undergo substantial remodeling in pregnancy.¹³ The uterine circulation supports fetoplacental gas and nutrient exchange¹⁴ and shifts toward a dilatory, low-resistance phenotype during pregnancy.¹⁵ This adaptation is accompanied by hypertrophic outward remodeling and increases in vessel length and diameter,¹⁶ while spiral artery remodeling shifts control of vascular resistance upstream to the radial and arcuate arteries, highlighting these vessels as key contributors.

In addition to the maternal uterine vasculature, the placenta contains distinct maternal and fetal vascular compartments. On the maternal side, spiral arteries deliver maternal blood into the intervillous space. On the fetal

Highlights

- Endothelial dysfunction is a central mechanism of preeclampsia, affecting both uterine and systemic vascular beds.
- Impaired trophoblast-endothelial-immune interactions disrupt spiral artery remodeling and uteroplacental perfusion.
- Placenta-derived circulating factors (angiogenic imbalance, inflammation, extracellular vesicles, and microRNAs) propagate endothelial injury.
- The uterine microcirculation remains understudied and represents a key target for future diagnostics and therapies.

side, umbilical arteries branch into chorionic plate vessels and then into the villous vascular tree, returning via chorionic veins to the umbilical vein. In preeclampsia, the placental vascular architecture can be distorted. Li et al¹⁷ reported substantial changes in the placental vasculature in mild and severe preeclampsia compared with uncomplicated controls. Preeclampsia has also been associated with reduced vascularization index and placental volume.¹⁸ These characteristics are clinically relevant as they can play an important role in the assessment of placental vascularization and perfusion in preeclamptic pregnancies. However, direct mechanistic data on placental vascular endothelial phenotype and function remain limited.

When compared with the nonpregnant state, uterine artery blood flow increases progressively across gestation, reaching >10-fold by term.^{19,20} This increase is driven by changes in maternal systemic hemodynamics, the impact of circulating factors (eg, estrogen and angiogenic factors such as VEGF [vascular endothelial growth factor]), and local molecular mediators (eg, placenta-derived vasoactive factors).¹⁶ A reduction in uteroplacental perfusion below the physiological reserve can compromise maternal-fetal gas exchange and fetal oxygenation,¹⁹ contributing to placental dysfunction, and, in some cases, fetal growth restriction,²¹ clinical features that commonly accompany preeclampsia.²² Vascular mechanisms underlying these deficits are attributed to impaired uterine spiral artery remodeling and an imbalance between the production of vasodilators and constrictors.²³

TROPHOBLAST-ENDOTHELIAL-IMMUNE CELL INTERACTIONS AND IMPAIRED UTERINE SPIRAL ARTERY REMODELING

A subset of women with preeclampsia exhibits shallow trophoblast invasion, which can impair uterine spiral artery remodeling.²⁴ Structurally, spiral arteries may be smaller in diameter and length compared with arteries from healthy pregnancies.²⁵ This limited diameter, not

reaching 200 to 500 μm , leads to decreased blood flow through the uterine circulation, which has been confirmed with Doppler ultrasound.²⁶ In addition, reduced spiral artery length may limit the extent of these vascular networks into the fetal space of the placenta. Preeclamptic placentas show acute atherosclerosis (atherosclerotic-like lesions) in the uteroplacental vasculature,²⁷ which has been attributed to impaired maternal-fetal immunologic crosstalk, including specific maternal killer immunoglobulin-like receptors and fetal human leukocyte antigen interactions.²⁸

During the early first trimester, extravillous trophoblasts migrate to the maternal decidual stroma.^{27,29} Interstitial extravillous trophoblasts infiltrate the decidua and interact with immune and decidual stromal cells, while endovascular extravillous trophoblasts invade the lumen of spiral arteries and form transient plugs that limit blood flow until ≈ 12 weeks of gestation.³⁰ This creates a low-flow environment at the maternal-fetal interface that supports placental development.^{29,31}

During migration into maternal spiral arteries, endovascular extravillous trophoblasts replace the endothelial layer and adopt an endothelial-like (pseudoendothelial) phenotype.³² Through this process, trophoblasts contribute to structural remodeling of the arterial wall, including disruption and loss of vascular smooth muscle.^{33,34} This transition is physiologically important because it converts spiral arteries into high-capacitance, low-resistance vessels with markedly reduced vasoreactivity, supporting increased uteroplacental perfusion during gestation.^{33,34} In parallel, extravillous trophoblasts can promote immune cell tolerance in part by inducing apoptosis of selected maternal immune cells.³⁰ These processes are facilitated in part by Fas-/FasL-mediated apoptosis, occurring within a tightly regulated cytokine and immune signaling environment at the maternal-fetal interface.³⁵

Although the exact mechanisms governing trophoblast-endothelial interactions have not been fully elucidated, transcriptomic profiling of extravillous trophoblast differentiation programs shows enrichment of cell adhesion pathways, including integrin-mediated signaling, consistent with an important role for integrins in trophoblast-endothelial interactions.³⁶ During their transition toward a pseudoendothelial (vascular adhesion) phenotype, extravillous trophoblasts downregulate epithelial-associated adhesion factors while upregulating endothelial-like molecules (eg, vascular endothelial cadherin, platelet endothelial cell adhesion molecule-1, and soluble VCAM-1 [vascular cell adhesion molecule-1]) and related adhesion programs.³⁷ In preeclampsia, however, this transition does not occur, and endovascular extravillous trophoblasts do not express these endothelial molecules, failing to assume an endothelial phenotype.²⁵

It is also important to note that trophoblast lineages beyond extravillous trophoblasts may contribute to endothelial dysfunction in preeclampsia. Syncytiotrophoblasts

form the villous surface of the placenta and mediate maternal-fetal exchange while adapting to changing circulatory demands across gestation.³⁸ In both early- and late-onset preeclampsia, a central shared feature is syncytiotrophoblast stress, which can arise not only from impaired oxygen delivery due to abnormal placentation and malperfusion but also from progressive placental cell spatial constraint later in gestation (eg, increasing villous crowding and compression of the intervillous space), accompanied by oxidative and inflammatory stress, and activation of cell death and stress pathways.³⁹ Stressed syncytiotrophoblasts can contribute to maternal endothelial dysfunction by promoting angiogenic imbalance, release of extracellular vesicles (EVs), and proinflammatory and procoagulant factors.³⁹ These downstream effectors will be discussed in subsequent sections of this review.

Some studies suggest that infiltration of spiral arteries by immune cells (eg, uterine natural killer cells and decidual macrophages) precedes and facilitates trophoblastic invasion, implicating nontrophoblastic contributors to spiral artery remodeling.⁴⁰ In pregnancies with high uterine artery pulsatility or resistance indices, including preeclampsia, evidence suggests impaired uterine immune-trophoblast signaling that may limit trophoblast invasion and spiral artery remodeling.⁴¹ Uterine natural killer cells are abundant in early pregnancy and are critical regulators of trophoblast invasion and uterine vascular remodeling through secretion of cytokines/chemokines and angiogenic factors and through crosstalk with extravillous trophoblasts.^{40,42,43} As part of this remodeling process, uterine natural killer cell-associated signaling contributes to disorganization and loss of vascular smooth muscle surrounding the spiral arteries, promoting the transition to a low-resistance, high-capacitance uteroplacental circulation.^{40,43} Consistent with this role, uterine natural killer cells are enriched in the decidua during the first trimester relative to the late gestation.⁴⁰ Together, these findings support the concept that impaired uterine spiral artery remodeling in preeclampsia reflects failed endothelial-trophoblast-immune crosstalk at the maternal-fetal interface.

Maternal blood supply to the placenta becomes functional near the end of the first trimester (≈ 11 – 12 weeks of human gestation)^{21,44–46} when trophoblast plugs dislodge, and blood flow to the fetoplacental unit is established.⁴⁷ In a subset of preeclamptic pregnancies, failure of spiral artery remodeling may contribute to persistent hypoperfusion and placental stress later in pregnancy, promoting aberrant placental and fetal development and the maternal syndrome of preeclampsia.²¹

Although systemic vascular dysfunction is a feature of both early- and late-onset preeclampsia, impaired uterine spiral artery remodeling is more strongly associated with the early-onset subtype, which is less common, typically more severe, and associated with worse maternal and

fetal outcomes.⁴⁸ This heterogeneity has been attributed, in part, to differences in immune regulation between the 2 subtypes. For example, targeted immune gene expression profiling and immunohistochemistry demonstrated that the placental innate immune pathways were altered in early-onset preeclampsia, whereas these alterations were not observed in late-onset placentas.⁴⁹

In early-onset preeclampsia, the authors reported decreased placental expression of TLR (toll-like receptor)-associated genes (eg, *TLR4* and *TLR1*), as well as complement-associated genes (*C7*, *C1QA*, *C2*, and *C1R*), while no changes in these pathways were observed in late-onset preeclampsia.⁴⁹ Because TLR and complement pathways can modulate endothelial activation, leukocyte adhesion, and barrier function, these placental immune features may have implications for maternal endothelial phenotype and therapeutic strategies.^{50–53}

Taken together, these observations support the concept that early- and late-onset preeclampsia may involve distinct placental immune-vascular states. Accordingly, endothelial-targeted therapies may require subtype-specific strategies, particularly because early-onset preeclampsia is often associated with uteroplacental deficiencies, whereas late-onset disease may reflect stronger contributions from maternal cardiometabolic or vascular vulnerabilities.

UTERINE ARTERY ENDOTHELIAL DYSFUNCTION

As spiral arteries remodel and their endothelial cells are replaced by trophoblasts at the beginning of pregnancy, changes in the vascular endothelium in upstream uterine arteries and their branches (ie, arcuate/radial/myometrial arteries) occur. During a healthy pregnancy, endothelial cell adaptations in these upstream vessels include growth/proliferation,⁵⁴ increased production of the vasodilators NO and PGI₂,^{55–57} enhanced gap-junctional communication involving connexins 43, 37, and 40,⁵⁸ and augmented contribution of endothelium-derived hyperpolarization to agonist-induced vasodilation.⁵⁹ In preeclampsia, however, adaptive uterine endothelial responses to pregnancy are significantly reduced. Impaired NO signaling in the uterine circulation may be driven by reduced endothelial NO synthase activity,⁶⁰ imbalance of L-arginine and tetrahydrobiopterin with increased arginase activity, and eNOS uncoupling due to oxidative stress.^{43,61}

In addition, reduced expression and activity of endothelial Ca²⁺-activated K⁺ channels, including IKCa (intermediate-conductance calcium-activated potassium channel) and SKCa (small-conductance calcium-activated potassium channel), have been reported in uterine arteries in preeclampsia⁶² and have been linked to oxidative stress and endoplasmic reticulum stress.⁶¹ Li

et al⁶¹ demonstrated that dilation of maternal small arteries differs between normal and preeclamptic pregnancies and involves contributions from IKCa-/SKCa-dependent pathways interacting with NO signaling. In chorionic plate resistance arteries, preeclampsia was associated with reduced eNOS and iNOS (inducible nitric oxide synthase) expression and decreased IKCa and SKCa channel density.⁶¹ Functionally, after NOS inhibition with L-NAME, compensatory IKCa-/SKCa-dependent dilation was diminished in arteries from preeclamptic pregnancies compared with controls, consistent with a role for endothelium-derived NO in modulating IKCa-/SKCa-mediated vasodilatation. Luksha et al⁶³ further support impaired hyperpolarization-dependent mechanisms in preeclampsia by demonstrating reduced endothelium-derived hyperpolarization-mediated dilatory responses through myoendothelial gap junctions in myometrial arteries from preeclamptic patients. In a complementary study, Choi et al⁶⁴ reported that the downregulation of endothelial IKCa channels in preeclampsia was associated with increased oxidative stress signaling, supporting a potential link between redox imbalance and reduced KCa channel-dependent endothelial function. Collectively, these findings suggest that impaired SKCa/IKCa channel function can reduce endothelial hyperpolarization and attenuate endothelium-derived hyperpolarization-mediated relaxation, contributing to reduced vasodilation and increased vascular resistance in preeclampsia.

CIRCULATING EFFECTORS OF ENDOTHELIAL CELL ACTIVATION AND POTENTIAL BIOMARKERS OF ENDOTHELIAL DYSFUNCTION

Flow-mediated dilation and other measures of systemic endothelium-dependent vasodilation are reduced in preeclampsia across vascular beds, and these impairments persist postpartum, supporting a systemic endothelial dysfunction phenotype.^{65–67} Mechanisms underlying these systemic vascular changes have been reviewed extensively^{23,68–71}; here, we focus on circulating mediators that may link uteroplacental dysfunction to maternal endothelial activation and injury.

In addition to vasodilatory dysfunction, endothelial cells show structural and barrier abnormalities in preeclampsia. The endothelial glycocalyx is disrupted, as indicated by increased shedding of syndecan-1 and hyaluronan.⁷² This disruption is consistent with increased endothelial permeability and a more adhesive endothelial phenotype, potentially amplifying vascular inflammation.

A growing body of evidence demonstrates that placental malperfusion and maladaptation of the uterine circulation contribute to maternal systemic dysfunction. This process is driven, in part, by placental release of biologically active mediators, including angiogenic factors,

inflammatory and adhesion molecules, oxidative stress-related mediators, EVs, and microRNA.^{39,73,74} In this section, we discuss major classes of circulating mediators that reflect and potentially propagate endothelial dysfunction.

Angiogenic Factors

Impaired trophoblast invasion is a unique feature of preeclampsia that is associated with the release of antiangiogenic and proapoptotic factors into the circulation and alterations in the oxygen tension in the uterine circulation.^{75–78} The increased release of antiangiogenic factors in preeclampsia is related to placental hypoperfusion resulting from reduced spiral artery remodeling and shallow trophoblast invasion. These circulating factors are vasoactive and directly affect maternal cardiovascular function. For instance, activation of VEGF receptor-2 (kinase domain-related receptor) by VEGF on endothelial cells induces production and release of vasodilatory NO and PGI₂⁷⁹; thus, reduced VEGF signaling may reduce NO and PGI₂ bioavailability and contribute to endothelial dysfunction. PIGF (placental growth factor) is a placenta-derived member of the VEGF family that supports angiogenesis during pregnancy. PIGF signals primarily through the VEGFR-1 (VEGF receptor-1/Flt-1 [fms-like tyrosine kinase 1]) expressed on endothelial and myeloid cells,⁸⁰ and contributes to placental vascular development, including expansion and branching of the vascular network in the placental villous tree.⁸¹ In preeclampsia, circulating concentrations of sFlt (soluble fms-like tyrosine kinase)-1 are increased, resulting in an angiogenic imbalance characterized by an elevated sFlt/PIGF ratio. sFlt-1 acts as a decoy receptor that binds and sequesters VEGF and PIGF, thereby inhibiting their interaction with endothelial receptors and blunting VEGF-dependent mitogenic and vasodilatory effects.^{82–86} Similarly, soluble endoglin, a placenta-derived soluble TGF- β (transforming growth factor-beta) coreceptor that is elevated in preeclampsia, disrupts TGF- β -dependent signaling in the endothelium, including reduced eNOS activation, thereby impairing vasodilation.⁸⁷

In pregnant women with preeclampsia, Taraseviciene et al⁸⁸ showed higher levels of sFlt-1, sFlt-1/PIGF ratio, lower levels of PIGF, and higher mean uterine artery pulsatility and resistance indices. In preclinical animal models, uteroplacental ischemia, induced by mechanically restricting uterine artery blood flow, is associated with increased circulating sFlt-1 and reduced MMPs (matrix metalloproteinases) and collagen deposition in uterine arteries.^{89,90} MMPs are involved in trophoblast invasion, uterine spiral artery remodeling, and angiogenesis⁹¹; thus, sFlt-1-induced MMP reduction could significantly impact these processes, leading to reduced uteroplacental vascularization. Infusion of sFlt-1 in healthy pregnant rats caused a similar phenotype,^{89,90} while treatment of

both animal models with PIGF reversed these vascular outcomes. These findings suggest a close association of angiogenic factors with changes in uterine artery blood flow, remodeling, and resistance in preeclamptic pregnancies.

Inflammatory Markers, Adhesion Molecules, and Oxidative Stress Mediators

Proinflammatory cytokines are increased in preeclampsia, whereas anti-inflammatory cytokines, IL (interleukin)-4 and IL-10, are decreased.^{92–94} Among these, TNF- α (tumor necrosis factor-alpha) is consistently elevated,^{95,96} and maternal plasma of soluble TNF receptor-2 increases before the clinical manifestation of preeclampsia.⁹⁷ TNF- α is strongly associated with endothelial dysfunction, possibly due to its effects on the NO pathway, including induction of mitochondrial reactive oxygen species production in human umbilical vein endothelial cells⁹⁸ and downregulation of eNOS mRNA expression and stability.^{99,100} In parallel, IL-6 is increased in cases of chronic hypertension with superimposed preeclampsia and in women at high risk for term preeclampsia,¹⁰¹ and it is associated with maternal endothelial dysfunction.

In addition to cytokines, other inflammatory biomarkers, including high-sensitivity C-reactive protein, desquimated endotheliocytes, and growth differentiation factor 15, are also increased in preeclampsia.^{102–104} Women with preeclampsia also have higher concentrations of serum VCAM-1, and these increases are significant in patients with chronic hypertension and superimposed preeclampsia.¹⁰⁵ Levels of soluble intercellular adhesion molecule-1 and E-selectin concentrations follow a similar pattern,¹⁰⁶ indicative of inflammation and endothelial dysfunction.

Markers of oxidative stress (eg, reactive oxygen species/oxidative metabolites and the lipid peroxidation product malondialdehyde) are elevated in preeclampsia,^{107,108} whereas indices of antioxidant capacity, including total antioxidant capacity and the ferric reducing ability of plasma, are reduced.¹⁰⁹ In some studies, pregnancies with eclampsia show higher malondialdehyde levels than those with preeclampsia, supporting an association between oxidative stress burden and disease severity.¹¹⁰ Mechanistically, oxidative stress can reduce NO bioavailability, promote vasoconstrictor signaling, and impair endothelial mitochondrial/redox homeostasis, thereby shifting the uterine circulation toward increased tone and resistance.¹¹¹ In agreement with this concept, lower systemic oxidative stress has been correlated with lower uterine artery pulsatility, suggesting a link between oxidative stress and uteroplacental perfusion.¹¹²

Direct associations between elevated markers of endothelial activation/inflammation and reduced NO bioavailability suggest that impaired NO signaling is a major contributor to endothelial dysfunction in preeclampsia.¹¹³

Higher levels of sVCAM-1 and sFlt-1 are associated with lower NO synthesis,¹¹⁴ and maternal systolic blood pressure is positively correlated with soluble E-selectin, sVCAM-1, and sFlt-1.¹¹⁴ Thus, proinflammatory factors likely regulate vascular function and represent therapeutic targets in complicated pregnancies.

EVs and microRNAs

Circulating EVs and microRNAs are dynamic components of intercellular communication in pregnancy and have been implicated in preeclampsia.¹¹⁵ EVs are released into the circulation from multiple sources, including placental trophoblast/syncytiotrophoblast, endothelial and immune cells, and maternal organs, and can be detected in blood, urine, and amniotic

fluid.¹¹⁶ In normal pregnancy, EVs and their cargo support adhesion of trophoblast cells and extracellular matrix remodeling during implantation, immune modulation to support tolerance of the growing fetus, vascular adaptation of spiral arteries and uterine vessels, and immune preparation for delivery.^{116–120} In preeclampsia, circulating EV profiles are consistently altered,^{121–124} most commonly characterized by increased EV abundance and shifts in cargo composition that favor proinflammatory,¹²⁴ antiangiogenic, oxidative stress, and endothelial-activating signaling,¹²⁵ linking EVs to systemic endothelial dysfunction (reviewed in detail in the study by Hula *et al*¹²⁶). However, the relative contribution of specific EV sources across normal pregnancy versus preeclampsia and the role of these EVs in the uterine circulation remain incompletely defined.

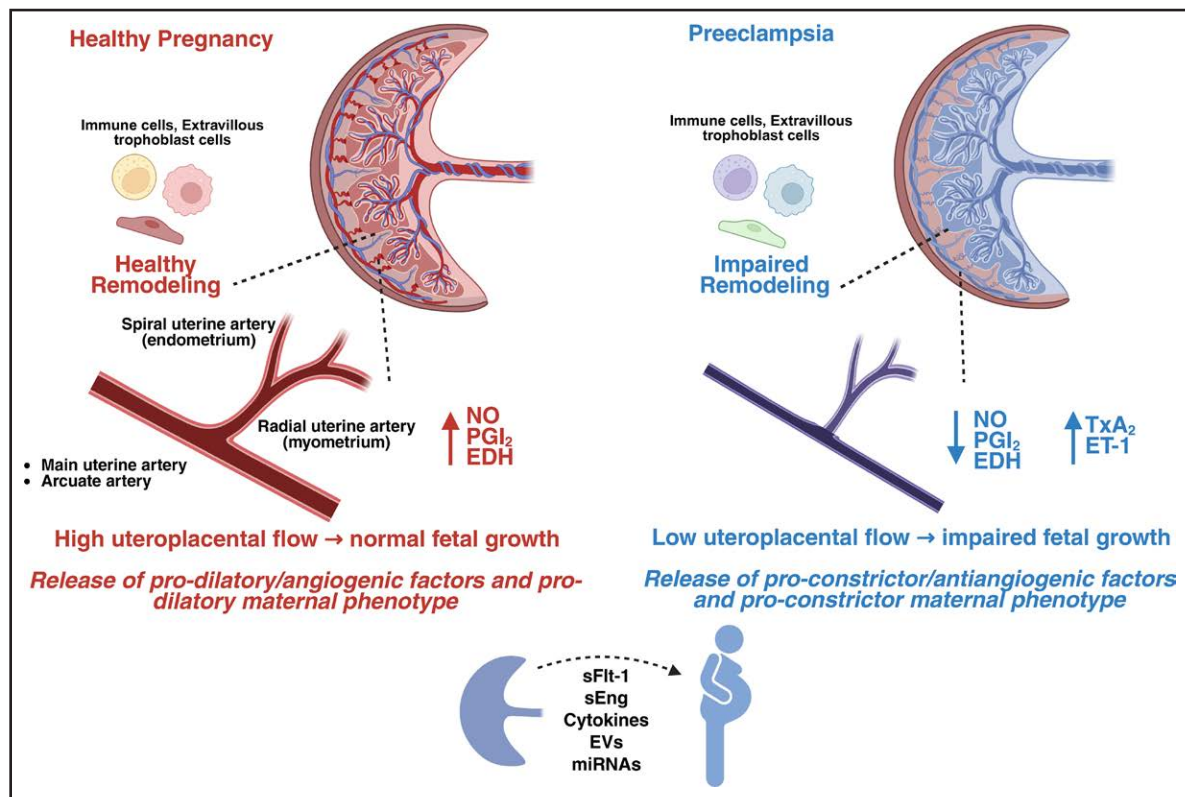


Figure. Conceptual overview of local uteroplacental vascular adaptations in healthy pregnancy vs preeclampsia and placenta-to-maternal circulating signals.

Left, Healthy pregnancy: the uterine circulation undergoes extensive structural and functional remodeling. Coordinated interactions among invading trophoblasts, and maternal endothelial and immune cells promote the transformation of uterine spiral arteries (uterine resistance vessels) into high-capacitance, low-resistance conduits, increasing uteroplacental perfusion. Upstream uterine arteries also remodel and exhibit reduced tone through enhanced endothelium-dependent vasodilator pathways, including NO, PGI₂ (prostacyclin 2), and endothelium-derived hyperpolarization (EDH). Collectively, these adaptations increase uterine blood flow, support normal fetal growth, and facilitate the release of prodilatory/angiogenic factors in the maternal circulation, where they support a prodilatory maternal phenotype. **Right,** Preeclampsia: spiral artery remodeling is impaired, and vessels retain tone and higher resistance, limiting placental perfusion. Upstream vessels fail to adequately remodel and shift toward a vasoconstrictor phenotype characterized by reduced NO/PGI₂/EDH signaling and increased constrictor influences (eg, thromboxane A₂ [TxA₂] and ET-1 [endothelin-1]). These maladaptations reduce uteroplacental blood flow and contribute to impaired fetal growth and the release of proconstrictor/antiangiogenic factors in the maternal circulation, where they promote a proconstrictor phenotype. **Bottom,** Placenta-maternal communication: placenta-derived extracellular vesicles (EVs) and their cargo (including microRNAs [miRNAs]), together with soluble circulating factors and cytokines, are shown as candidate mediators linking placental stress to maternal endothelial/vascular responses. Dotted lines indicate magnified views of the placental bed/spiral artery site, while arterial branches illustrate upstream uterine vessels (ie, arcuate, radial, and main uterine arteries). sEng indicates soluble endoglin; and sFlt-1, soluble fms-like tyrosine kinase-1.

Circulating microRNAs are stable posttranscriptional regulators,¹²⁷ and aberrant microRNA profiles in preeclampsia impact pathways relevant to trophoblast invasion and uterine vascular adaptations.¹²⁸ Some upstream stressors that may promote changes in circulating microRNAs include hypoxia, accumulation of reactive oxygen species, and inflammation, altered trophoblast differentiation, and changes in EV vesicle biogenesis and cargo, all characteristics of preeclampsia.^{126,129} Changes in circulating microRNAs may, therefore, reflect maternal organ dysfunction, including endothelial dysfunction and injury.^{130,131} For example, microRNAs targeting VEGF are associated with maternal vascular dysfunction in preeclampsia.¹³² Experimental overexpression of miR-210 in pregnant mice induces hypertension, endothelial dysfunction, and proteinuria¹³³ and inhibits trophoblast invasion.¹³⁴ Plasma miR-125b, which is upregulated in the 12th to 13th weeks of gestation in women who later developed preeclampsia, inhibits trophoblast invasion *in vitro* and reduces tube formation in human umbilical vein endothelial cells.¹³⁵ Together, these findings suggest that altered microRNA expression contributes to both placental and vascular dysfunction in preeclampsia. EV-associated microRNAs also differ between preeclampsia and healthy pregnancies^{136,137}; compartment-specific

patterns may help distinguish maternal and placental contributions, and guide targeted interventions.

CONCLUSIONS AND OPPORTUNITIES FOR FUTURE RESEARCH

This review emphasizes that preeclampsia is linked to endothelial dysfunction, with the uterine circulation representing a vascular bed with unique vulnerabilities. We posit that the uterine vasculature and its microcirculation contribute to the pathogenesis of the disease and reflect a patient's endothelial vulnerability. Uterine artery maladaptation likely bridges placental stress and systemic maternal endothelial dysfunction, integrating maternal and fetal signals. These signals are reflected in the maternal circulation through changes in circulating molecules such as cytokines, angiogenic, and vasoactive factors, which circulate either freely or within EVs carrying molecular signatures of organs under stress, including the uteroplacental unit (Figure).

Despite advances in systemic vascular assessments, endothelial function in patients with preeclampsia has been primarily studied in peripheral conduit and resistance vessels, whereas the uterine circulation, which undergoes the most dramatic vascular

Table. Major Knowledge Gaps and Key Future Questions in Uteroplacental Endothelial Dysfunction in Preeclampsia

Theme/area	What we know	Major gap	Key questions	Approaches/tools to address
Uteroplacental endothelial phenotype	Peripheral endothelial dysfunction is documented; uterine circulation is less studied.	Direct data on uterine/placental endothelial phenotypes (microvasculature vs microvasculature)	Which endothelial subtypes change? What pathways define maladaptation?	Placental microvascular endothelial cell isolation, spatial/single-cell profiling, glycocalyx, and pericyte assays
Preclinical models	Mechanical perfusion-restriction and pathway-focused models recapitulate select preeclampsia-like symptoms.	Limited modeling of early placental/spiral artery remodeling and endothelial phenotypic transitions	Which models best match uteroplacental endothelial dysfunction? How to standardize end points?	Harmonized uterine artery end points, placental microvascular readouts, and cross-model comparisons
Temporary trajectory of uterine endothelial dysfunction across gestation	Clinical syndrome manifests late; early pathogenesis begins earlier.	Longitudinal uteroplacental endothelial function measures are scarce.	When does dysfunction begin? Is it reversible?	Longitudinal biomarker panels and uterine artery Doppler with circulating endothelial marker panels
Clinical subtypes	Heterogeneity exists; immune/angiogenic profiles vary.	Uteroplacental endothelial phenotypes mapped poorly by subtype/endotype	Do endothelial pathways differ by subtype? What is shared vs distinct?	Stratified cohorts, angiogenic/inflammatory profiling, and machine learning clustering
Maternal vs fetal placental vascular compartments	Distinct maternal intervillous vs fetal villous vascular trees.	Compartment-specific endothelial dysfunction mechanisms are unclear.	Which side drives circulating signals? How do they interact?	Compartment-focused sampling, imaging/casting, and microvascular functional assays
Circulating mediators and EV cargo	Cytokines/angiogenic factors/EVs change in preeclampsia.	Placental origin of circulating mediators and whether they are causal or reflective is often unclear.	What combination best predicts disease? What are causal effectors?	Multomics EV profiling, causal mediation analyses, and endothelial bioassays
Noninvasive assessment tools	Doppler is widely used; endothelial function tests are limited in pregnancy.	Lack of pregnancy-safe, endothelial-specific functional readouts for the uteroplacental vascular bed	Can we develop/validate noninvasive endothelial metrics?	Imaging and circulating endothelial activation panels; digital biomarkers
Population heterogeneity and social drivers of health	Preeclampsia risk and outcomes vary by population and context.	Most mechanistic studies include small and nonrepresentative cohorts.	How do social drivers of health and environment modify uteroplacental endothelial vulnerability?	Diverse cohorts, harmonized meta-data, and artificial intelligence models incorporating social drivers of health and environmental triggers

EV indicates extracellular vesicle.

adaptations during pregnancy, remains understudied. In the Table, we summarize major gaps and key future questions in uteroplacental endothelial dysfunction in preeclampsia. Future work should extend beyond conduit arteries and trophoblast cells to include pericytes, the endothelial glycocalyx, and comprehensive profiling of uterine endothelial subtypes in patients with and without preeclampsia. Defining the mechanisms that trigger and facilitate uterine endothelial transitions during healthy pregnancy could set the foundation for understanding how maladaptations can be reversed in preeclampsia.

Preclinical animal models provide a critical platform to study interorgan communication, integrative physiological responses, and maternal vascular adaptations in preeclampsia.¹³⁸ However, commonly used animal models vary in their relevance to uteroplacental vascular biology.¹³⁹ Animal models with mechanical restriction of uterine arteries (eg, reduced uteroplacental perfusion models and RUPP/sRUPP) enable controlled reductions in uteroplacental blood flow and facilitate downstream assessment of uterine artery function; however, they are typically implemented in mid-to-late gestation and, therefore, do not capture early immune maladaptation, abnormal trophoblast invasion, and spiral artery remodeling.¹³⁹ In addition, the phenotype is not reversible because the mechanical restriction is permanent. Models that focus on specific molecular pathways (eg, NOS inhibition, sFlt-1, cytokines, and TLR stimulation) allow testing of cause-and-effect relationships between specific mediators and endothelial dysfunction but often focus on a single pathway in a healthy background and do not capture the evolution of the uteroplacental circulation during pregnancy.¹³⁹

Efforts to map the repertoire of circulating factors that bridge maternal and fetal signals of uterine stress could improve prognosis and allow early detection of abnormalities that lead to preeclampsia. Critical gaps that require attention include temporal changes in uterine endothelial dysfunction and its associations with systemic maternal endothelial dysfunction during pregnancy and postpartum, differences in endothelial phenotypes between clinical subtypes of preeclampsia, and the limitations introduced by small, convenience samples that do not reflect population heterogeneity.

A key challenge in studying the uterine circulation is the limited access to human tissue and the lack of non-invasive assessments to test endothelial function during pregnancy. Progress requires new strategies, beyond traditional assessments of vascular function and models of preeclampsia. Integrating high-dimensional vascular and circulating biomarker data with artificial intelligence approaches that also incorporate environmental and social drivers of vascular health offers a path to leverage existing human and experimental data, enhance

understanding of uterine endothelial biology, and ultimately inform targeted therapies in preeclampsia.

ARTICLE INFORMATION

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ChatGPT (GPT-5.2, OpenAI) was used solely for grammar, syntax, and typographic editing in portions of this manuscript. It was not used to generate scientific content or draft original text. The authors are solely responsible for the conception, writing, and final content of this work.

Author Contributions

S.M. Tucker, N. Hula, and J.J. Gardner drafted the manuscript. S. Gouloupoulou edited and revised the manuscript and prepared the figure.

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None.

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